

Question 1 of 63



Choose 1 option.

Refer to the code below:

```
01 let a = 'a';  
02 let b;  
03 // b = a;  
04 console.log(b);
```

What is displayed when the code executes?

- ☐ A. ReferenceError: b is not defined
- ☒ B. undefined
- ☐ C. a
- ☐ D. null

Question 2 of 63



Choose 1 option.

A developer wants to use a module called `DatePrettyPrint`. This module exports one default function called `printDate()`.

How can a developer import and use the `printDate()` function?

- ☐ A. `import DatePrettyPrint from '/path/DatePrettyPrint.js';`
`DatePrettyPrint.printDate();`
- ☐ B. `import DatePrettyPrint() from '/path/DatePrettyPrint.js';`
`printDate();`
- ☒ C. `import printDate from '/path/DatePrettyPrint.js';`
`printDate();`
- ☐ D. `import printDate from '/path/DatePrettyPrint.js';`
`DatePrettyPrint.printDate();`

Question 3 of 63

Choose 1 option.

Refer to the code below:

```
01 function Person() {  
02   this.firstName = 'John';  
03 }  
04  
05 Person.prototype = {  
06   job: x => 'Developer'  
07 };  
08  
09 const myFather = new Person();  
10 const result = myFather.firstName + ' ' + myFather.job();
```

What is the value of `result` after line 10 executes?

☐ A. John undefined

What is the value of `result` after line 10 executes?

- ☐ A. John undefined
- ☒ B. Error: `myFather.job` is not a function
- ☐ C. John Developer
- ☐ D. undefined Developer

Question 4 of 63

Choose 1 option.

Refer to the code below:

```
01 let timedFunction = () => {  
02   console.log('Timer called.');
```

```
03 };  
04  
05 let timerId = setTimeout(timedFunction, 1000);
```

Which statement allows a developer to cancel the scheduled timed function?

- ☐ A. `removeTimeout(timedFunction);`
- ☒ B. `clearTimeout(timerId);`
- ☐ C. `clearTimeout(timedFunction);`
- ☐ D. `removeTimeout(timerId);`

Question 5 of 63

Choose 2 options.

Refer to the following array:

```
let arr1 = [1, 2, 3, 4, 5];
```

Which two lines of code result in a second array, `arr2`, being created such that `arr2` is a reference to `arr1`?

- ☒ A. `let arr2 = arr1.sort();`
- ☐ B. `let arr2 = Array.from(arr1);`
- ☐ C. `let arr2 = arr1.slice(0, 5);`
- ☒ D. `let arr2 = arr1;`

Question 6 of 63



Choose 1 option.

A developer creates a class that represents a news story based on the requirements that a Story should have a body, author, and view count. The code is shown below:

```
01 class Story {  
02     // Insert code here  
03     this.body = body;  
04     this.author = author;  
05     this.viewCount = viewCount;  
06 }  
07 }
```

Which statement should be inserted in the placeholder on line 02 to allow for a variable to be set to a new instance of a Story with the three attributes correctly populated?

- ☒ A. `constructor(body, author, viewCount) {`
- ☐ B. `function Story(body, author, viewCount) {`

- ☒ A. constructor(body, author, viewCount) {
- ☐ B. function Story(body, author, viewCount) {
- ☐ C. constructor() {
- ☐ D. super (body, author, viewCount) {

Question 7 of 63

Choose 1 option.

Refer to the code below:

```
01 <html lang="en">
02   <table onclick="console.log('Table log');">
03     <tr id="row1">
04       <td>Click me!</td>
05     </tr>
06   </table>
07   <script>
08     function printMessage(event) {
09       console.log('Row log');
10     }
11
12     let elem = document.getElementById('row1');
13     elem.addEventListener('click', printMessage, false);
14   </script>
15 </html>
```

Which code change should be made for the console to log only `Row log` when 'Click me!' is clicked?

```
15 </html>
```

Which code change should be made for the console to log only Row 1 log when 'Click me!' is clicked?

- ☒ A. Add `event.stopPropagation();` to `printMessage` function.
- ☐ B. Add `event.removeEventListener();` to `printMessage` function.
- ☐ C. Add `event.removeEventListener();` to `window.onload` event handler.
- ☐ D. Add `event.stopPropagation();` to `window.onload` event handler.

Question 8 of 63



Choose 3 options.

A developer creates a generic function to log custom messages in the console. To do this, the function below is implemented.

```
01 function logStatus(status){  
02   console.log('Answer goes here*/('Item status is: %s', status);  
03 }
```

Which three console logging methods allow the use of string substitution in line 02?

☐ A. assert

☒ B. info

☒ C. error

☒ D. log

Question 9 of 63



Choose 1 option.

Refer to the following code:

```
01 <html lang="en">
02   <div class="outerDiv">
03     <button class="myButton">Click me!</button>
04   </div>
05   <script>
06     function displayInnerMessage(ev) {
07       console.log('Inner message. ');
08     }
09     function displayOuterMessage(ev) {
10       console.log('Outer message. ');
11     }
12
13     window.onload = (event) => {
14       document.querySelector('.outerDiv')
15         .addEventListener('click', displayOuterMessage, true);
16       document.querySelector('.myButton')
17         .addEventListener('click', displayInnerMessage, true);
18     };
```

```
16     document.querySelector('.myButton')
17         .addEventListener('click', displayInnerMessage, true);
18 };
19 </script>
20 </html>
```

What will the console show when the button is clicked?

- ☐ A. > Inner message.
- ☐ B. > Inner message.
> Outer message.
- ☒ C. > Outer message.
> Inner message.
- ☐ D. > Outer message.

Question 10 of 63



Choose 1 option.

Universal Containers recently launched its new landing page to host a crowd-funding campaign. The page uses an external library to display some third-party ads. Once the page is fully loaded, it creates more than 50 new HTML items placed randomly inside the DOM, like the one in the code below:

```
<!-- This is an ad -->
<div class="ad-library-item ad-hidden" onload="myFunction()" >
  
</div>
```

All the elements include the same `ad-library-item` class. They are hidden by default, and they are randomly displayed while the user navigates through the page.

Tired of all the ads, what can the developer do to temporarily and quickly remove them?

- ☒ A. Use the browser console to execute a script that removes all the elements containing the class `ad-library-item`.

Tired of all the ads, what can the developer do to temporarily and quickly remove them?

- ☒ A. Use the browser console to execute a script that removes all the elements containing the class `ad-library-item`.
- ☐ B. Use the DOM inspector to prevent the `load` event to be fired.
- ☐ C. Use the browser console to execute a script that prevents the `load` event to be fired.

Question 11 of 63

Choose 1 option.

Refer to the code below:

```
01 function changeValue(param) {  
02   param = 5;  
03 }  
04 let a = 10;  
05 changeValue(a);  
06  
07 let b = a;  
08 const result = a + ' - ' + b;
```

What is the value of result when the code executes?

- ☐ A. 10-May
- ☒ B. 10-Oct
- ☐ C. 5-May

Question 12 of 63

Choose 2 options.

A developer has a `fizzbuzz` function that, when passed in a number, returns the following:

- `'fizz'` if the number is divisible by 3.
- `'buzz'` if the number is divisible by 5.
- `'fizzbuzz'` if the number is divisible by both 3 and 5.
- empty string if the number is divisible by neither 3 or 5.

Which two test cases properly test scenarios for the `fizzbuzz` function?

☒ A. `let res = fizzbuzz(15);`
`console.assert(res === 'fizzbuzz');`

☒ B. `let res = fizzbuzz(Infinity);`
`console.assert(res === '');`

☐ C. `let res = fizzbuzz(3);`
`console.assert(res === 'buzz');`

☒ A. `let res = fizzbuzz(15);`
`console.assert(res === 'fizzbuzz');`

☒ B. `let res = fizzbuzz(Infinity);`
`console.assert(res === '');`

☐ C. `let res = fizzbuzz(3);`
`console.assert(res === 'buzz');`

☐ D. `let res = fizzbuzz(NaN);`
`console.assert(isNaN(res));`

Question 13 of 63

Choose 1 option.

Refer to the code below:

```
flag();  
anotherFlag();
```

```
function flag() {  
  console.log('flag');  
}
```

```
const anotherFlag = () => {  
  console.log('another flag');  
}
```

What is result of the code block?

- ☐ A. The console logs 'flag' and 'another flag'.
- ☒ B. The console logs only 'flag'.

What is result of the code block?

- ☐ A. The console logs 'flag' and 'another flag'.
- ☒ B. The console logs only 'flag'.
- ☐ C. The console logs 'flag' and then an error is thrown.
- ☐ D. An error is thrown.

Question 14 of 63

Choose 1 option.

Given the HTML below:

```
<div>
  <div id="row-uc">Universal Containers</div>
  <div id="row-as">Applied Shipping</div>
  <div id="row-bt">Burlington Textiles</div>
</div>
```

Which statement adds the `priority-account` CSS class to the Universal Containers row?

- ☐ A. `document.querySelectorAll('#row-uc').classList.add('priority-account');`
- ☐ B. `document.getElementById('row-uc').addClass('priority-account');`
- ☒ C. `document.querySelector('#row-uc').classList.add('priority-account');`
- ☐ D. `document.querySelector('#row-uc').classes.push('priority-account');`

Question 15 of 63

Choose 1 option.

Considering type coercion, What does the following expression evaluate to?

```
true + 3 + '100' + null
```

☐ A. '3100null'

☐ B. 104

☐ C. 4100

☒ D. '4100null'

Question 16 of 63

Choose 2 options.

A developer uses a parsed JSON string to work with user information as shown in the code block below:

```
01 const userInfo = {  
02     "id" : "user-01",  
03     "email" : "user01@universalcontainers.demo",  
04     "age" : 25  
05 };
```

Which two options access the age attribute in the object?

- ☐ A. `userInfo.get('age');`
- ☒ B. `userInfo('age');`
- ☐ C. `const {age} = userInfo;`
- ☒ D. `userInfo.age;`

Question 17 of 63

Choose 1 option.

Given the code below:

```
01 setTimeout(() => {  
02   console.log(1);  
03 }, 1100);  
04 console.log(2);  
05 new Promise((resolve, reject) => {  
06   setTimeout(() => {  
07     reject(console.log(3));  
08   }, 1000);  
09 }).catch(() => {  
10   console.log(4);  
11 });  
12 console.log(5);
```

What is logged to the console?

☐ A. 2 5 1 3 4


```
12 console.log(5);
```

What is logged to the console?

☐ A. 2 5 1 3 4

☐ B. 1 2 5 3 4

☒ C. 2 5 3 4 1

☐ D. 1 2 3 4 5

Question 18 of 63

Choose 1 option.

Refer to the code below:

```
01 function execute() {  
02   return new Promise((resolve, reject) => reject());  
03 }  
04 let promise = execute();  
05  
06 promise  
07   .then(() => console.log('Resolved1'))  
08   .then(() => console.log('Resolved2'))  
09   .then(() => console.log('Resolved3'))  
10   .catch(() => console.log('Rejected'))  
11   .then(() => console.log('Resolved4'));
```

What is the result when the `Promise` in the `execute` function is rejected?

What is the result when the `Promise` in the `execute` function is rejected?

☐ A. `Resolved1 Resolved2 Resolved3 Rejected Resolved4`

☐ B. `Rejected`

☒ C. `Rejected Resolved4`

☐ D. `Resolved1 Resolved2 Resolved3 Resolved4`

Question 19 of 63

Choose 1 option.

Refer to the HTML below:

```
<div id="main">
  <ul>
    <li>Leo</li>
    <li>Tony</li>
    <li>Tiger</li>
  </ul>
</div>
```

Which JavaScript statement results in changing "Tony" to "Mr. T."?

- ☐ A. `document.querySelectorAll('#main #TONY').innerHTML = 'Mr. T.';`
- ☒ B. `document.querySelector('#main li:nth-child(2)').innerHTML = 'Mr. T.';`
- ☐ C. `document.querySelectorAll('#main li.Tony').innerHTML = 'Mr. T.';`

- ☐ A. `document.querySelectorAll('#main #TONY').innerHTML = 'Mr. T.';`
- ☒ B. `document.querySelector('#main li:nth-child(2)').innerHTML = 'Mr. T.';`
- ☐ C. `document.querySelectorAll('#main li.Tony').innerHTML = 'Mr. T.';`
- ☐ D. `document.querySelector('#main li:second-child').innerHTML = 'Mr. T.';`

Question 19 of 63

Choose 1 option.

Refer to the HTML below:

```
<div id="main">
  <ul>
    <li>Leo</li>
    <li>Tony</li>
    <li>Tiger</li>
  </ul>
</div>
```

Which JavaScript statement results in changing "Tony" to "Mr. T."?

- ☐ A. `document.querySelectorAll('#main #TONY').innerHTML = 'Mr. T.';`
- ☒ B. `document.querySelector('#main li:nth-child(2)').innerHTML = 'Mr. T.';`
- ☐ C. `document.querySelectorAll('#main li.Tony').innerHTML = 'Mr. T.';`

Question 20 of 63

Choose 1 option.

Refer to the code below:

```
let textValue = '1984';
```

Which code segment shows a correct way to convert this string to an integer?

- ☐ A. `let numberValue = (Number)textValue;`
- ☒ B. `let numberValue = Number(textValue);`
- ☐ C. `let numberValue = textValue.toInteger();`
- ☐ D. `let numberValue = Integer(textValue);`

Question 21 of 63



Choose 1 option.

A developer is setting up a new Node.js server with a client library that is built using events and callbacks.

The library:

- Will establish a web socket connection and handle receipt of messages to the server
- Will be imported with `require`, and made available with a variable called `ws`.

The developer also wants to add error logging if a connection fails.

Given this information, which code segment shows the correct way to set up a client with two events that listen at execution time?

☐ A.

```
04 ws.connect(() => {  
05   console.log('Connected to client');  
06 }).catch((error) => {  
07   console.log('ERROR', error);  
08 });
```


Choose 1 option.

```
06 }).catch((error) => {  
07   console.log('ERROR', error);  
08 });
```

☒ B. 04 ws.on('connect', () => {
05 console.log('Connected to client');
06 });
07
08 ws.on('error', (error) => {
09 console.log('ERROR', error);
10 });

☐ C. 04 try {
05 ws.connect(() => {
06 console.log('Connected to client');
07 });
08 } catch(error) {
09 console.log('ERROR', error);
10 }

☐ C. 04 try {
05 ws.connect(() => {
06 console.log('Connected to client');
07 });
08 } catch(error) {
09 console.log('ERROR', error);
10 }

☐ D. 04 ws.on('connect', () => {
05 console.log('Connected to client');
06
07 ws.on('error', (error) => {
08 console.log('ERROR', error);
09 });
10 });

Question 22 of 63



Choose 2 options.

Given two expressions `var1` and `var2`, what are two valid ways to return the logical AND of the two expressions and ensure it is data type `Boolean`?

- ☒ A. `Boolean(var1 && var2)`
- ☐ B. `var1.toBoolean() && var2.toBoolean()`
- ☐ C. `var1 && var2`
- ☒ D. `Boolean(var1) && Boolean(var2)`

Question 23 of 63

Choose 1 option.

Refer to the following code block:

```
01 class Student {
02     constructor(name) {
03         this.name = name;
04     }
05
06     takeTest() {
07         console.log(`${this.name} got 70% on test.`);
08     }
09 }
10
11 class BetterStudent extends Student {
12     constructor(name) {
13         super(name);
14         this.name = 'Better student ' + name;
15     }
16     takeTest() {
17         console.log(`${this.name} got 100% on test.`);
18     }
```

```
16   takeTest() {  
17       console.log(`${this.name} got 100% on test.`);  
18   }  
19 }  
20  
21 let student = new BetterStudent('Jackie');  
22 student.takeTest();
```

What is the console output?

- ☐ A. > Uncaught ReferenceError
- ☐ B. > Better student Jackie got 70% on test.
- ☒ C. > Better student Jackie got 100% on test.
- ☐ D. > Jackie got 70% on test.

Question 24 of 63



Choose 1 option.

A developer has the following code:

```
01 let x = object.value;
02
03 try {
04   handleObjectValue(x);
05 } catch(error) {
06   handleError(error);
07 }
```

The developer has a `getNextValue()` function to execute after `handleObjectValue()`. Also, the developer wants to execute `getNextValue()` regardless of whether an error occurs.

How can the developer change the code to ensure this behavior?

How can the developer change the code to ensure this behavior?

☒ A. 03 try {
04 handleObjectValue(x);
05 getNextValue();
06 } catch(error) {
07 handleError(error);
08 }

☐ B. 03 try {
04 handleObjectValue(x);
05 } catch(error) {
06 handleError(error);
07 } finally {
08 getNextValue();
09 }

☐ C. 03 try {
04 handleObjectValue(x);
05 } catch(error) {
06 if (error) {
07 handleError(error);
08 } else {
09 getNextValue();
10 }
11 }

☐ D. 03 try {
04 handleObjectValue(x);
05 } catch(error) {
06 getNextValue();
07 handleError(error);
08 }

Question 25 of 63



Choose 1 option.

A developer writes the code below to return a message to a user attempting to register a new username. If the username is available, a variable named `msg` is declared and assigned a value on line 03.

```
01  function getAvailabilityMessage(item) {  
02      if (getAvailability(item)) {  
03          var msg = "Username available";  
04      }  
05      return msg;  
06  }
```

What is the value of `msg` when `getAvailabilityMessage("newUserName")` is executed and `getAvailability("newUserName")` returns `true`?

- ☐ A. "msg is not defined"
- ☐ B. undefined
- ☒ C. "Username available"

☐ A. "msg is not defined"

☐ B. undefined

☒ C. "Username available"

☐ D. "newUserName"

Question 26 of 63

Choose 3 options.

Refer to the following code:

```
let sampleText = 'The quick brown fox jumps';
```

A developer needs to determine if a certain substring is part of a string.

Which three expressions return true for the given substring?

- ☒ A. `sampleText.substring('fox');`
- ☐ B. `sampleText.includes('Fox', 3);`
- ☒ C. `sampleText.includes('quick', 4);`
- ☒ D. `sampleText.indexOf('quick') !== -1;`
- ☐ E. `sampleText.includes('fox');`

Question 27 of 63

Choose 1 option.

A developer has the function, shown below, that is called when a page loads.

```
function onLoad() {  
    console.log("Page has loaded!");  
}
```

Where can the developer see the log statement after loading the page in the browser?

- ☐ A. On the terminal console running the web server
- ☐ B. On the webpage console log
- ☒ C. On the browser JavaScript console
- ☐ D. In the browser performance tools log

Question 28 of 63

Choose 3 options.

A developer implements a function that adds a few values.

```
01 function sum(num) {  
02   if (num === undefined) {  
03     num = 0;  
04   }  
05   return function(num2, num3) {  
06     if (num3 === undefined) {  
07       num3 = 0;  
08     }  
09     return num + num2 + num3;  
10   }  
11 }
```

Which three options can the developer invoke for this function to get a return value of 10?

Which three options can the developer invoke for this function to get a return value of 10?

☒ A. `sum() (10)`

☐ B. `sum(5, 5) ()`

☐ C. `sum(10) ()`

☒ D. `sum() (5, 5)`

☒ E. `sum(5) (5)`

Question 29 of 63

Choose 3 options.

Refer to the code declarations below:

```
let str1 = 'Java';  
let str2 = 'Script';
```

Which three expressions return the string JavaScript?

- ☒ A. `'${str1}${str2}'`;
- ☐ B. `concat(str1, str2)`;
- ☐ C. `str1.join(str2)`;
- ☒ D. `str1 + str2`;
- ☒ E. `str1.concat(str2)`;

Question 30 of 63



Choose 1 option.

A developer needs to debug a Node.js web server because a runtime error keeps occurring at one of the endpoints.

The developer wants to test the endpoint on a local machine and make the request against a local server to look at the behavior. In the source code, the `server.js` file will start the server. The developer wants to debug the Node.js server only using the terminal.

Which command can the developer use to open the CLI debugger in their current terminal window?

- ☐ A. `node start inspect server.js`
- ☐ B. `node server.js --inspect`
- ☐ C. `node -i server.js`
- ☒ D. `node inspect server.js`

Question 31 of 63

Choose 2 options.

The developer wants to test the array shown:

```
const arr = Array(5).fill(0);
```

Which two tests are the most accurate for this array?

- ☐ A. `console.assert(arr[0] === 0 && arr[arr.length] === 0);`
- ☐ B. `console.assert(arr.length > 0);`
- ☒ C. `arr.forEach(elem => console.assert(elem === 0));`
- ☒ D. `console.assert(arr.length === 5);`

Question 32 of 63

Choose 1 option.

Which statement can a developer apply to increment the browser's navigation history without a page refresh?

- ☒ A. `window.history.pushState(newStateObject, '', null);`
- ☐ B. `window.history.state.push(newStateObject);`
- ☐ C. `window.history.pushState(newStateObject);`
- ☐ D. `window.history.replaceState(newStateObject, '', null);`



Question 33 of 63

Choose 2 options.

A developer wrote the following code to test a `sum3` function that takes in an array of numbers and returns the sum of the first three numbers in the array, and the test passes.

```
01 let res = sum3([1, 4, 1]);  
02 console.assert(res === 6);  
03  
04 res = sum3([1, 5, 0, 5]);  
05 console.assert(res === 6);
```

A different developer made changes to the behavior of `sum3` to instead sum only the first two numbers present in the array.

Which two results occur when running this test on the updated `sum3` function?

- ☒ A. The line 02 assertion passes.
- ☐ B. The line 05 assertion passes.
- ☐ C. The line 02 assertion fails.

Which two results occur when running this test on the updated skills function?

- ☒ A. The line 02 assertion passes.
- ☐ B. The line 05 assertion passes.
- ☐ C. The line 02 assertion fails.
- ☒ D. The line 05 assertion fails.

Question 34 of 63

Choose 1 option.

Refer to the following code:

```
01 function test(val) {  
02   if (val === undefined) {  
03     return 'Undefined value!';  
04   }  
05   if (val === null) {  
06     return 'Null value!';  
07   }  
08   return val;  
09 }  
10  
11 let x;  
12  
13 test(x);
```

What is returned by the function call on line 13?

What is returned by the function call on line 13?

- ☒ A. 'Undefined value!'
- ☐ B. undefined
- ☐ C. 'Null value!'
- ☐ D. Line 13 throws an error.

Question 35 of 63



Choose 1 option.

A developer wants to create a simple image upload in the browser using the File API. The HTML is below:

```
<input type="file" onchange="previewFile()">
<img src="" height="200" alt="Image preview..."/>
```

The JavaScript portion is:

```
01 function previewFile() {
02   const preview = document.querySelector('img');
03   const file = document.querySelector('input[type=file]').files[0];
04   // line 4 code
05   reader.addEventListener("load", () => {
06     preview.src = reader.result;
07   }, false);
08   //line 8 code
09 }
```

In lines 04 and 08, which code allows the user to select an image from their local computer, and to display the image in the browser?

In lines 04 and 08, which code allows the user to select an image from their local computer, and to display the image in the browser?

- ☒ A.

```
04 const reader = new FileReader();  
08 if (file) reader.readAsDataURL(file);
```
- ☐ B.

```
04 const reader = new File();  
08 if (file) URL.createObjectURL(file);
```
- ☐ C.

```
04 const reader = new File();  
08 if (file) reader.readAsDataURL(file);
```
- ☐ D.

```
04 const reader = new FileReader();  
08 if (file) URL.createObjectURL(file);
```


Question 36 of 63

Choose 1 option.

Refer to the code below:

```
01 console.log('Start');  
02 Promise.resolve('Success').then(function(value) {  
03   console.log('Success');  
04 });  
05 console.log('End');
```

What is the output after the code executes successfully?

- ☐ A. Success
Start
End
- ☐ B. Start
Success
End

End



C. Start

End

Success



D. End

Start

Success

Question 37 of 63



Choose 1 option.

A developer creates a simple webpage with an input field. When a user enters text in the input field and clicks the button, the actual value of the field must be displayed in the console.

Here is the HTML file content:

```
<input type="text" value="Hello" name="input">
<button type="button">Display</button>
```

The developer wrote the JavaScript code below:

```
01 const button = document.querySelector('button');
02 button.addEventListener('click', () => {
03   const input = document.querySelector('input');
04   console.log(input.getAttribute('value'));
05 });
```

When the user clicks the button, the output is always "Hello".

What needs to be done to make this code work as expected?

```
05 });
```

When the user clicks the button, the output is always "Hello".

What needs to be done to make this code work as expected?

- ☐ A. Replace line 02 with `button.addEventListener("onclick", function() {`
- ☐ B. Replace line 03 with `const input = document.getElementById('input');`
- ☐ C. Replace line 02 with `button.addCallback("click", function() {`
- ☒ D. Replace line 04 with `console.log(input.value);`

Question 38 of 63

Choose 1 option.

Given the code below:

```
01  function myFunction() {  
02      a = 5;  
03      var b = 1;  
04  }  
05  
06  myFunction();  
07  
08  console.log(a);  
09  console.log(b);
```

What is the expected output?

- ☒ A. Line 08 outputs the variable, but line 09 throws an error.
- ☐ B. Both lines 08 and 09 are executed, but the values outputted are undefined.
- ☐ C. Line 08 throws an error, therefore line 09 is never executed.
- ☐ D. Both lines 08 and 09 are executed, and the variables are outputted.

Question 39 of 63

Choose 1 option.

Which statement accurately describes an aspect of promises?

- ☒ A. Arguments for the callback function passed to `.then()` are optional.
- ☐ B. In a `.then()` function, returning results is not necessary since callbacks will catch the result of a previous promise.
- ☐ C. `.then()` manipulates and returns the original promise.
- ☐ D. `.then()` cannot be added after a `catch`.

Question 40 of 63

Choose 2 options.

Refer to the following code that imports a module named `Utils`:

```
01 import {foo,bar} from '/path/Utils.js';  
02 foo();  
03 bar();
```

Which two implementations of `Utils.js` export `foo` and `bar` such that the code above runs without error?

☒ A.

```
const foo = () => { return 'foo'; }  
const bar = () => { return 'bar'; }  
export default foo, bar;
```

☒ B.

```
export default class {  
  foo() { return 'foo'; }  
  bar() { return 'bar'; }  
}
```



```
foo() { return 'foo'; }  
bar() { return 'bar'; }  
}
```

☐ C. `const foo = () => { return 'foo'; }
const bar = () => { return 'bar'; }
export {foo, bar}`

☐ D. `// FooUtils.js and BarUtils.js exist
import {foo} from '/path/FooUtils.js';
import {bar} from '/path/BarUtils.js';
export {foo, bar}`

Question 41 of 63



Choose 1 option.

At Universal Containers, every team has its own way of copying JavaScript objects. The code snippet shows an implementation from one team:

```
01 function Person() {  
02   this.firstName = "John";  
03   this.lastName = "Doe";  
04   this.name = () => {  
05     console.log(`Hello ${this.firstName} ${this.lastName}`);  
06   }  
07 }  
08  
09 const john = new Person();  
10 const dan = JSON.parse(JSON.stringify(john));  
11 dan.firstName = 'Dan';  
12 dan.name();
```

What is the output of the code execution?

What is the output of the code execution?

- ☐ A. TypeError: Assignment to constant variable
- ☐ B. Hello Dan Doe
- ☒ C. TypeError: dan.name is not a function
- ☐ D. Hello John Doe

Question 42 of 63

Choose 1 option.

Refer to the following object.

```
01 const dog = {  
02   firstName: 'Beau',  
03   lastName: 'Boo',  
04   get fullName() {  
05     return this.firstName + ' ' + this.lastName;  
06   }  
07 };
```

How can a developer access the fullName property for dog?

- ☒ A. dog.fullName
- ☐ B. dog.get.fullName
- ☐ C. dog.function.fullName()

Question 43 of 63



Choose 3 options.

A developer is trying to convince management that their team will benefit from using Node.js for a backend server that they are going to create. The server will be a web server that handles API requests from a website that the team has already built using HTML, CSS, and JavaScript.

Which three benefits of Node.js can the developer use to persuade their manager?

- ☐ A. Ensures stability with one major release every few years.
- ☒ B. Installs with its own package manager to install and manage third-party libraries.
- ☒ C. Uses non-blocking functionality for performant request handling.
- ☐ D. Performs a static analysis on code before execution to look for runtime errors.
- ☒ E. Executes server-side JavaScript code to avoid learning a new language.

Question 44 of 63

Choose 1 option.

A developer has an `ErrorHandler` module that contains multiple functions.

What kind of export should be leveraged so that multiple functions can be used?

- ☐ A. multi
- ☐ B. default
- ☒ C. named
- ☐ D. all

Question 45 of 63



Choose 2 options.

A developer is leading the creation of a new web server for their team that will fulfill API requests from an existing client.

The team wants a web server that runs on Node.js, and they want to use the new web framework Minimalist.js. The lead developer wants to advocate for a more seasoned back-end framework that already has a community around it.

Which two frameworks could the lead developer advocate for?

- ☒ A. Angular
- ☐ B. Next.js
- ☒ C. Gatsby
- ☐ D. Nest

Question 46 of 63

Choose 1 option.

Given the code below:

```
const delay = async delay => {  
  return new Promise((resolve, reject) => {  
    setTimeout(resolve, delay);  
  });  
};
```

```
const callDelay = async () => {  
  const yup = await delay(1000);  
  console.log(1);  
};
```

```
console.log(2);  
callDelay();  
console.log(3);
```

What is logged to the console?


```
};
```

```
console.log(2);  
callDelay();  
console.log(3);
```

What is logged to the console?

☐ A. 1 2 3

☐ B. 2 1 3

☒ C. 2 3 1

☐ D. 1 3 2

Question 47 of 63

Choose 2 options.

Which two actions can be done using the JavaScript browser console?

- ☐ A. View the security cookies.
- ☒ B. Run code that's not related to the page.
- ☒ C. Change the DOM and the JavaScript code of the page.
- ☐ D. Display a report showing the performance of a page.

Question 48 of 63



Choose 1 option.

A developer is required to write a function that calculates the sum of elements in an array but is getting undefined every time the code is executed. The developer needs to find what is missing in the code below.

```
01 const sumFunction = arr => {  
02   return arr.reduce((result, current) => {  
03     //  
04     result += current;  
05     //  
06   }, 10);  
07 };
```

Which line replacement makes the code work as expected?

- ☐ A. 02 return arr.forEach((result, current) => {
- ☒ B. 05 return result;
- ☐ C. 03 if(arr.length == 0) { return 0; }

Which line replacement makes the code work as expected?

- ☐ A. 02 `return arr.forEach((result, current) => {`
- ☒ B. 05 `return result;`
- ☐ C. 03 `if(arr.length == 0) { return 0; }`
- ☐ D. 04 `current = result + current;`

Question 49 of 63



Choose 2 options.

A developer is creating a simple webpage with a button. When a user clicks this button for the first time, a message is displayed.

The developer wrote the JavaScript code below, but something is missing. The message gets displayed every time a user clicks the button, instead of just the first time.

```
01 function listen(event) {  
02  
03   alert('Hey! I am John Doe');  
04  
05 }  
06 button.addEventListener('click', listen);
```

Which two code lines make this code work as required?

- ☐ A. On line 02, use `event.first` to test if it is the first execution.
- ☒ B. On line 06, add an option called `once` to `button.addEventListener()`.

- ☐ A. On line 02, use `event.first` to test if it is the first execution.
- ☒ B. On line 06, add an option called `once` to `button.addEventListener()`.
- ☒ C. On line 04, use `button.removeEventListener('click', listen);`
- ☐ D. On line 04, use `event.stopPropagation();`

Question 50 of 63

Choose 1 option.

Refer to the code below:

```
01 let car1 = new Promise((_ , reject) =>
02   setTimeout(reject, 2000, "Car 1 crashed in"));
03 let car2 = new Promise(resolve => setTimeout(resolve, 1500, "Car 2 completed"));
04 let car3 = new Promise(resolve => setTimeout(resolve, 3000, "Car 3 completed"));
05
06 Promise.race([car1, car2, car3])
07   .then(value => {
08     let result = `${value} the race.`;
09   })
10   .catch(err => {
11     console.log("Race is cancelled.", err);
12   });
```

What is the value of result when Promise.race executes?

What is the value of `result` when `Promise.race` executes?

- ☐ A. Car 3 completed the race.
- ☒ B. Car 2 completed the race.
- ☐ C. Car 1 crashed in the race.
- ☐ D. Race is cancelled.

Question 51 of 63



Choose 1 option.

A developer publishes a new version of a package with new features that do not break backward compatibility. The previous version number was 1.1.3.

Following semantic versioning format, what should the new package version number be?

- ☒ A. 1.2.0
- ☐ B. 1.1.4
- ☐ C. 2.0.0
- ☐ D. 1.2.3

Question 52 of 63

Choose 1 option.

Refer to the code below:

```
01 let first = 'Who';
02 let second = 'What';
03 try {
04     try {
05         throw new Error('Sad trombone');
06     } catch (err) {
07         first = 'Why';
08     } finally {
09         second = 'When';
10     }
11 } catch (err) {
12     second = 'Where';
13 }
```

What are the values for `first` and `second` once the code executes?

What are the values for `first` and `second` once the code executes?

- ☐ A. `first` is `Why` and `second` is `Where`.
- ☐ B. `first` is `Who` and `second` is `When`.
- ☐ C. `first` is `Who` and `second` is `Where`.
- ☒ D. `first` is `Why` and `second` is `When`.

Question 53 of 63

Choose 1 option.

Refer to the following code:

```
01 let obj = {  
02   foo: 1,  
03   bar: 2  
04 }  
05 let output = [];  
06  
07 for (let something in obj) {  
08   output.push(something);  
09 }  
10  
11 console.log(output);
```

What is the output of line 11?

What is the output of line 11?

☐ A. ["bar", "foo"]

☒ B. ["foo", "bar"]

☐ C. [1, 2]

☐ D. ["foo:1", "bar:2"]

Question 54 of 63

Choose 1 option.

Given the following code:

```
01 counter = 0;
02 const logCounter = () => {
03   console.log(counter);
04 };
05 logCounter();
06 setTimeout(logCounter, 1100);
07 setInterval(() => {
08   counter++;
09   logCounter();
10 }, 1000);
```

What is logged by the first four log statements?

☒ A. 0 1 1 2

☐ B. 0 1 2 3

☒ A. 0 1 1 2

☐ B. 0 1 2 3

☐ C. 0 0 1 2

☐ D. 0 1 2 2

Question 55 of 63

Choose 2 options.

Refer to the code below:

```
01 const exec = (item, delay) =>
02   new Promise(resolve => setTimeout(() => resolve(item), delay));
03
04 async function runParallel() {
05   const [result1, result2, result3] = await Promise.all(
06     [exec('x', '100'), exec('y', '500'), exec('z', '100')]
07   );
08   return `parallel is done: ${result1}${result2}${result3}`;
09 }
```

Which two statements correctly execute the `runParallel()` function?

Which two statements correctly execute the `runParallel()` function?

☒ A. `runParallel().done(function(data) {
 return data;
});`

☐ B. `async runParallel().then(data);`

☒ C. `runParallel().then(function(data) {
 return data;
});`

☐ D. `runParallel().then(data);`

Question 56 of 63

Choose 1 option.

Refer to the code below:

```
01 function foo() {  
02   const a = 2;  
03   function bar() {  
04     console.log(a);  
05   }  
06   return bar;  
07 }
```

Why does the function `bar` have access to variable `a`?

- ☐ A. Prototype chain
- ☐ B. Hoisting
- ☒ C. Outer function's scope

☐ B. Hoisting

☒ C. Outer function's scope

☐ D. Inner function's scope

Question 57 of 63

Choose 1 option.

Refer to the following code:

```
01 class Vehicle {
02     constructor(plate) {
03         this.plate = plate;
04     }
05 }
06
07 class Truck extends Vehicle {
08     constructor(plate, weight){
09         //Missing code
10         this.weight = weight;
11     }
12     displayWeight() {
13         console.log(`The truck ${this.plate} has a weight of ${this.weight} lb.`);
14     }
15 }
16
17 let myTruck = new Truck('123AB', 5000);
18 myTruck.displayWeight();
```

```
16  
17 let myTruck = new Truck('123AB', 5000);  
18 myTruck.displayWeight();
```

Which statement should be added to line 09 for the code to display 'The truck 123AB has a weight of 5000 lb.'?

- ☐ A. `super.plate = plate;`
- ☐ B. `this.plate = plate;`
- ☒ C. `super(plate);`
- ☐ D. `Vehicle.plate = plate;`

Question 58 of 63

Choose 2 options.

Given a value, which two options can a developer use to detect if the value is NaN?

☒ A. `Object.is(value, NaN)`

☐ B. `value == NaN`

☐ C. `value === Number.NaN`

☒ D. `isNaN(value)`

Question 59 of 63

Choose 3 options.

Refer to the code:

```
01 function Animal(size, type) {  
02   this.size = size || 'small';  
03   this.type = type || 'Animal';  
04   this.canTalk = false;  
05 }  
06  
07 let Pet = function(size, type, name, owner) {  
08   Animal.call(this, size, type);  
09   this.name = name;  
10   this.owner = owner;  
11 }  
12  
13 Pet.prototype = Object.create(Animal.prototype);  
14 let pet1 = new Pet();
```

Given the code above, which three properties are set for pet1?

Given the code above, which three properties are set for pet1?

- ☐ A. size
- ☒ B. canTalk
- ☒ C. type
- ☒ D. name
- ☐ E. owner

Question 60 of 63



Choose 2 options.

A developer is asked to fix some bugs reported by users. To do that, the developer adds a breakpoint for debugging.

```
01 function Car(maxSpeed, color){  
02     this.maxSpeed = maxSpeed;  
03     this.color = color;  
04 }  
05 let carSpeed = document.getElementById('carSpeed');  
06 debugger;  
07 let fourWheels = new Car(carSpeed.value, 'red');
```

When the code execution stops at the breakpoint on line 06, which two types of information are available in the browser console?

- ☒ A. A variable displaying the number of instances created for the Car object
- ☐ B. The information stored in the window.localStorage property
- ☐ C. The values of the carSpeed and fourWheels variables

- ☒ A. A variable displaying the number of instances created for the Car object
- ☐ B. The information stored in the window.localStorage property
- ☐ C. The values of the carSpeed and fourWheels variables
- ☒ D. The style, event listeners and other attributes applied to the carSpeed DOM element

Question 61 of 63

Choose 2 options.

Refer to the following code:

```
01 function Tiger() {  
02   this.type = 'Cat';  
03   this.size = 'large';  
04 }  
05  
06 let tony = new Tiger();  
07 tony.roar = () => {  
08   console.log('They\'re great!');  
09 };  
10  
11 function Lion() {  
12   this.type = 'Cat';  
13   this.size = 'large';  
14 }  
15  
16 let leo = new Lion();  
17 // Insert code here  
18 leo.roar();
```

☐ A. `Object.assign(leo, Tiger);`

☐ B. `leo.prototype.roar = () => { console.log('They\'re pretty good!'); };`

☒ C. `leo.roar = () => { console.log('They\'re pretty good!'); };`

☒ D. `Object.assign(leo, tony);`

Question 62 of 63

Choose 1 option.

Refer to the code below:

```
01 const myFunction = arr => {  
02   return arr.reduce((result, current) => {  
03     return result + current;  
04   }, 5);  
05 }
```

What is the output of this function when called with an empty array?

- ☐ A. Returns NaN
- ☐ B. Returns 0
- ☒ C. Returns 5
- ☐ D. Returns Infinity

Question 63 of 63



Choose 1 option.

A developer removes the HTML `class` attribute from the checkout button, so now it is simply:

```
<button>Checkout</button>
```

There is a test to verify the existence of the checkout button, however it looks for a button with `class="blue"`. The test fails because no such button is found.

Which type of test category describes this test?

- ☐ A. False positive
- ☐ B. True negative
- ☐ C. True positive
- ☒ D. False negative